

TrES-3b

R-2694

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190609 · created 2026-08-31T23:29:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458643.92233 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.033
Transit depth (Rp/Rs)^2	3.53 +/- 1.24 [%]
Orbital Inclination (inc)	81.26 +/- 1.12
Airmass coefficient 1 (a1)	0.969 +/- 0.013
Airmass coefficient 2 (a2)	0.03 +/- 0.021
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#2 - [566, 426]
Optimal Aperture	4.65
Optimal Annulus	11.04
Transit Duration (day)	0.05 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.188 $\pm$ 0.033 vs 0.16309 (deviation 0.4 σ)
Duration fit vs published	0.05 d vs 0.0567 d (deviation 0.23 σ)
Mid-time O-C	-4.5 min
Depth SNR	3.0
Residual scatter	1.38 %

Light curve

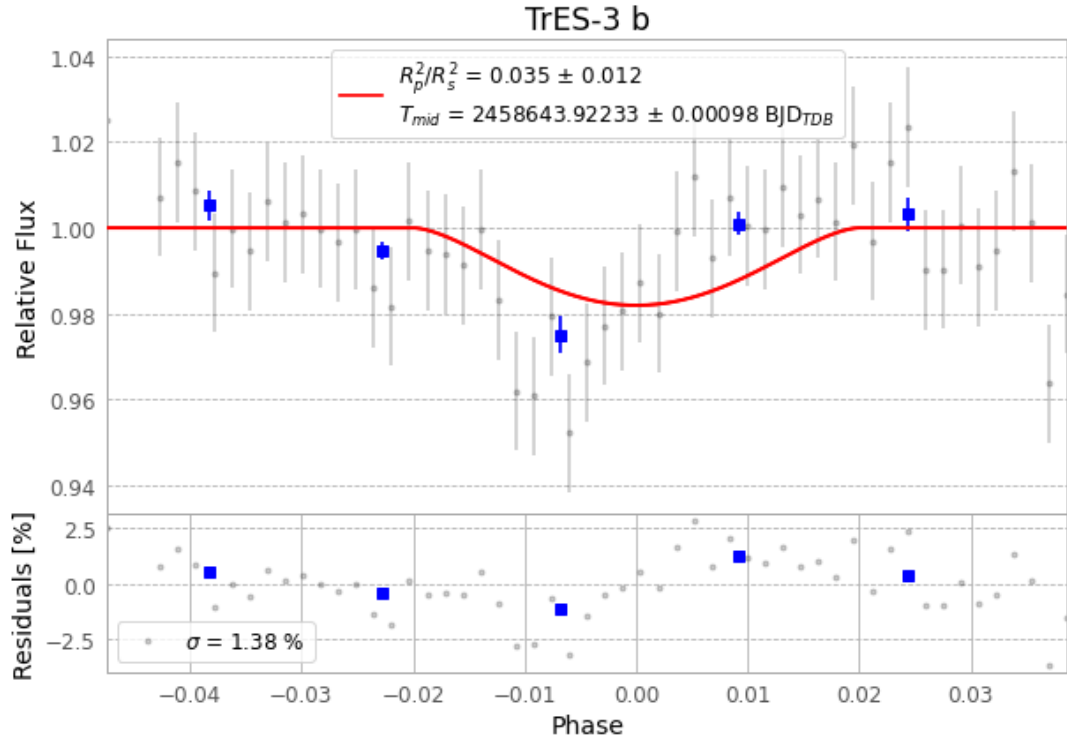


Figure 1 — FinalLightCurve_TrES-3 b_2019-06-09.png (SHA-256 aa6636bc4e4ad94f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d91c81a3bbc77f72...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), TRES-3190609083312.FITS → TRES-3190609111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190609.log (15a663ce504f...), FinalLightCurve_TrES-3 b_2019-06-09.png (aa6636bc4e4a...), FinalLightCurve_TrES-3 b_2019-06-09.csv (c6ab21039673...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 23:29Z.